

EDUCATION

University of Nebraska Omaha, USA	Jan 2025 — Dec 2026 (Expected)
<i>M.S. Computer Science (GPA: 4.0)</i> - Courseworks: Design and Analysis of Algorithms, Artificial Intelligence, Database Management System, Advanced Operating System, Software Specifications & Design, Cloud Computing	
Institute of Engineering, Pulchowk Campus, Tribhuvan University, Nepal	2018 — 2023
<i>B.E. Computer Engineering (GPA: 3.72)</i> - Courseworks: Probability & Statistics, Calculus, Artificial Intelligence, Software Specifications, Data Mining, Big Data Technologies, Data Structures, Object Oriented Programming, Computer Networks, Theory of Computation, Computer Architecture	

EXPERIENCE

CIST Graduate Research Assistant	Jan 2025 — Present
<i>University of Nebraska Omaha, Omaha, USA</i> Designed and implemented extensions to a declarative programming language by introducing functional syntax and custom operators, using parser construction and AST transformations to enable new expressive capabilities in Answer Set Programming. Built a Python-based compilation pipeline using the Clingo API to parse custom rules, extend solver AST nodes, and transform functional programs into standard ASP and back, enabling compatibility with existing solvers while preserving higher-level abstractions.	
Associate Software Engineer	Jun 2023 — Aug 2024
<i>GrowByData Services, Lalitpur, Nepal</i> Built and integrated scalable data ingestion and tracking services by developing API-driven integrations with Google Analytics, Google Ads, and Meta Ads, improving observability of user behavior and supporting data-driven decision-making for e-commerce clients. Developed end-to-end ETL and analytics workflows by extracting data from multiple backend services, cleaning and transforming high-volume datasets in Python, and delivering integrated dashboards and business intelligence insights to support customer-facing decision-making.	
Game Development Intern	Feb 2023 — Jun 2023
<i>Avana Game Studios, Dallas, TX, USA (Remote from Nepal)</i> Developed interactive game features in Unity using C#, created 3D assets in Blender and implemented a guitar simulation to map finger strumming to MIDI notes with audio validation against real instrument samples.	

SKILLS

Programming Languages: Python, C, C++, Java, SQL, JavaScript
Frameworks & Tools: Numpy, Pandas, Matplotlib, Scikit-learn, RESTful APIs, Git, Power BI, Snowflake, Linux
Machine Learning/Deep Learning: NLP, Computer Vision, LLM, Statistical Analysis, ETL, Data Visualization
Web Development: Django, Rest APIs, Flask, Fastapi, Streamlit, HTML, CSS, Bootstrap, React
Database & Deployment: MySQL, MongoDB, Docker, AWS, Azure, CI/CD
Soft Skills: Problem solving, Good communication, Teamwork, Leadership, Time management
Communication: English, Nepali, Hindi

PUBLICATIONS

- D.C. N. & Pandey, R., et al. (2023). *Improved Pre-processing Techniques for Mobile-based OCR Systems*. In Proceedings of the International Conference on Technologies for Computer Electrical, Electronics & Communication (ICT-CEEL) 2023, (pp. 15-22), Khwopa.

PROJECT

XAI User Study on LLM Integration with ML Systems 	2025
Built a web-based Explainable AI system integrating classical ML models with an LLM-driven chatbot to explain medical recommendations, and designed a backend data pipeline to log user interactions and study outcomes for quantitative analysis.	
Smart Attendance System 	2023
Built a real-time attendance system that extracts facial features on-device and transmits embeddings to the server, reducing network load while enabling secure, proximity-based attendance verification.	
Easy Read 	2022
Developed a reading aid for visually impaired users by converting images to speech using OCR and TTS, applying segmentation and noise filtering to improve text recognition accuracy by 50%.	

HONORS AND AWARDS

- Recipient of the \$5,000 **Graduate Research & Creative Activity (GRACA)** grant for summer research at the University of Nebraska Omaha.
- Batch Topper Award in Bachelor's 1st, 5th and 7th semesters** for achieving the highest grade in my program.
- First Prize Winner** in openMIS-DRG Datathon 2022, organized by Center for Applied Research and Development (CARD), Institute of Engineering (IOE), Tribhuvan University and the Asia eHealth Information Network (AeHIN), Philippines.